experiment-7

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA
```

```
df = pd.read_csv('/content/Wholesale customers data.csv')
```

```
df.drop_duplicates(inplace=True)
df.dropna(inplace=True)
```

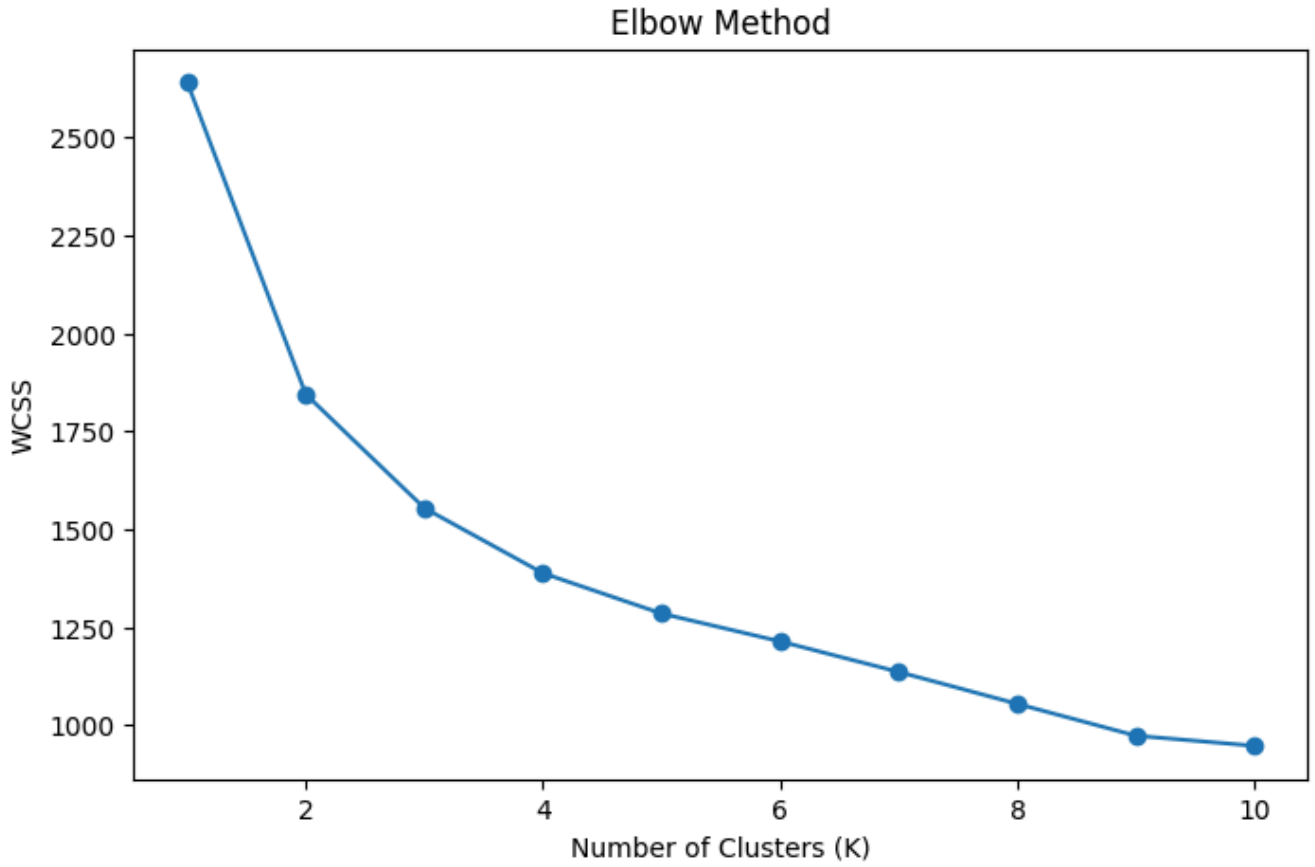
```
features = ['Fresh', 'Milk', 'Grocery', 'Frozen', 'Detergents_Paper', 'Delicassen']
X = df[features]
```

```
X_log = np.log1p(X)
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_log)
```

```
wcss = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
```

```
plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('WCSS')
plt.title('Elbow Method')
plt.show()
```




```
silhouette_scores = []
for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42)
    labels = kmeans.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    silhouette_scores.append(score)
```

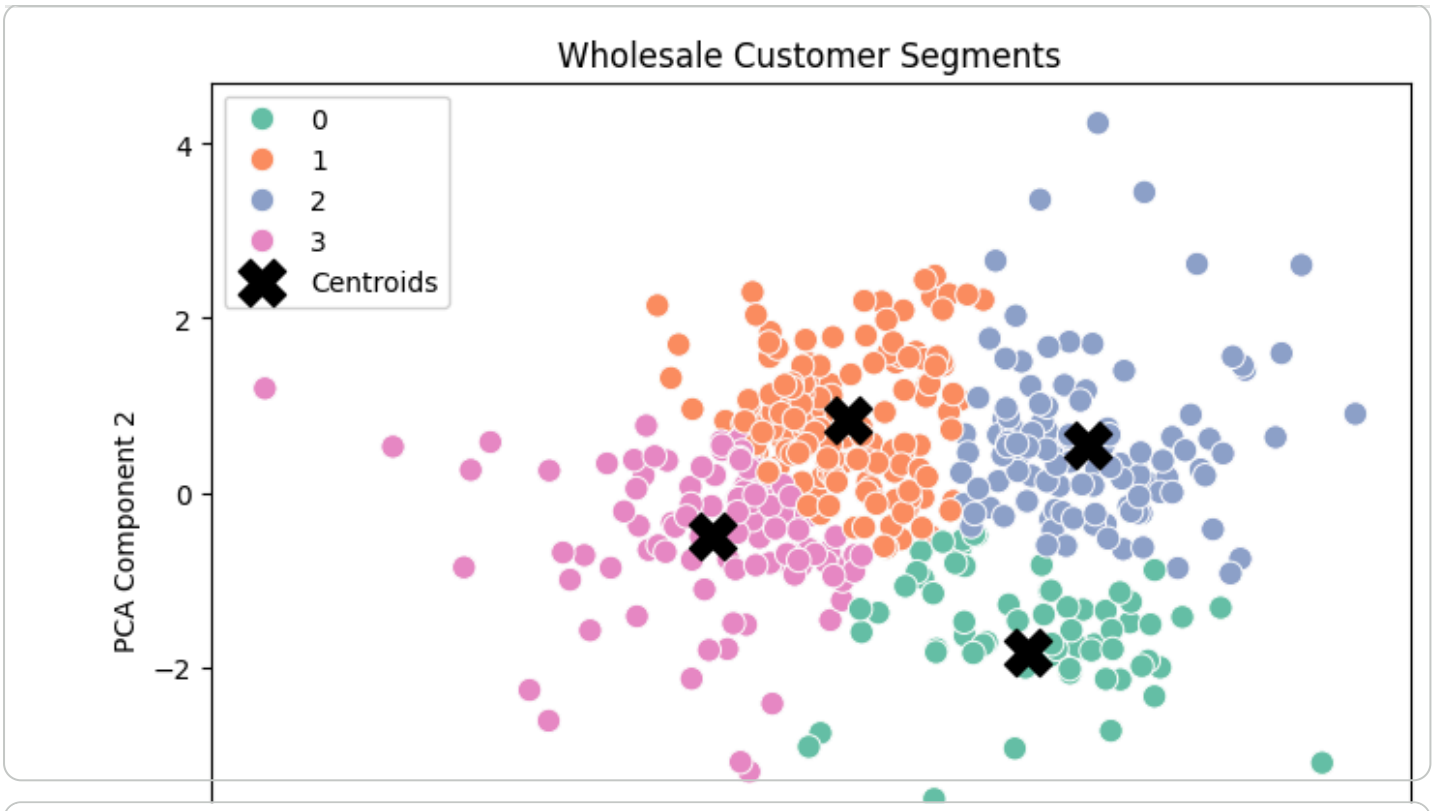
```
plt.figure(figsize=(8, 5))
plt.plot(range(2, 11), silhouette_scores, marker='o', color='green')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Silhouette Score')
plt.title('Silhouette Score for Different K')
plt.show()
```




```
optimal_k = 4
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42)
df['Cluster'] = kmeans.fit_predict(X_scaled)
```

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
df['PCA1'] = X_pca[:, 0]
df['PCA2'] = X_pca[:, 1]
```

```
plt.figure(figsize=(8, 6))
sns.scatterplot(x=df['PCA1'], y=df['PCA2'], hue=df['Cluster'], palette='Set2', s=80)
centroids_pca = pca.transform(kmeans.cluster_centers_)
plt.scatter(centroids_pca[:, 0], centroids_pca[:, 1], s=300, c='black', marker='X', la
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.title('Wholesale Customer Segments')
plt.legend()
plt.show()
```




```
cluster_summary = df.groupby('Cluster')[features].mean()
print(cluster_summary)
```

```
                Fresh          Milk       Grocery       Frozen  \
Cluster
0         2893.528571   6809.671429  11777.414286   597.214286
1        17531.896552   3493.413793   3851.951724  5027.965517
2        12703.000000  12307.756522  16522.939130  2942.904348
3         9769.218182   1379.481818   1958.836364  2203.236364

         Detergents_Paper   Delicassen
Cluster
```